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**INSTITUTE OF SCIENCE AND TECHNOLOGY
GODAWARI COLLEGE, ITAHARI**

A Project Proposal on NEXL: ELEVATE YOUR LEARNING

Submitted to
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*In partial fulfilment of the requirements for the degree of Bachelor of Science in Computer
Science and Information Technology (B.Sc. CSIT)*

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Declaration

We hereby declare that the project entitled “NEXL: Elevate Your Learning”, submitted to the Department of Computer Science and Information Technology, Tribhuvan University, is our original work and has been carried out by our group.

All sources of information and references used in the preparation of this project have been duly acknowledged. This project has not been submitted previously for any degree or academic award in this or any other institution.

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Abstract

The NEXL:Elevate Your Learning platform is designed to enhance the learning experience by providing an interactive and accessible web-based educational system. Traditional learning environments often face challenges such as limited accessibility, lack of personalized learning paths, and reduced student engagement. NEXL aims to overcome these challenges by offering a centralized platform where learners can enroll in courses, access structured learning materials, participate in quizzes, track progress, and earn achievement badges.

The system is developed using the MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring efficient data management, responsive user interfaces, and secure handling of user information. Key features of the platform include course management, user authentication, progress tracking, quiz and assessment modules, badge-based gamification, and administrative control for content management. NEXL provides a user-friendly, reliable, and scalable solution that supports both learners and educators, improving engagement, motivation, and overall learning effectiveness.

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1 Introduction

1.1 Introduction

The integration of digital technologies has transformed education, with Learning Management Systems (LMS) playing a central role. An LMS is a software platform for administering, delivering, tracking, and managing educational content and learner progress [1].

The COVID-19 pandemic accelerated the shift to online learning worldwide, including Nepal, where institutions adopted platforms like Zoom and Google Meet to continue education [5]. Studies show high LMS adoption: approximately 83% of students have internet access, with most using smartphones for learning (around 63% rely primarily on smartphones), yet they face significant challenges such as unreliable connectivity, irregular power supply, and limited digital skills [5, 6].

Modern LMS platforms incorporate AI personalization, multimedia support, communication tools, assessments, and recommendations to enhance engagement [4].

This project develops a web-based LMS tailored for Nepal, integrating advanced features to address current limitations and improve online education quality.

1.2 Problem Statement

Online education in Nepal has grown rapidly, but its effectiveness and sustainability are limited by financial, technical, and accessibility challenges [5]. High fees from international payment gateways, poor integration of local payment methods, and lack of flexible monetization tools make course creation and enrollment difficult. Additionally, most platforms lack interactive features, personalized recommendations, and multimedia support, reducing student engagement and retention [4]. This project aims to develop a web-based Learning Management System that integrates local payments, supports affordable course monetization, and offers interactive features to enhance the online learning experience for both teachers and students.

1.3 Objectives

The primary objective of this project is to design and develop a feature-rich, web-based Learning Management System that enhances accessibility, interactivity, and personalization in online education.

The specific objectives include implementing secure payment integration with coupon-based discounts, providing robust support for video recording, storage, AI-powered sum-

marization, and live streaming, enabling real-time group chat along with AI-driven course recommendations, remedial suggestions, and performance badges. The system will also support comprehensive teacher tools for course creation, pricing management, quiz and assignment design, and student features such as easy enrollment, course search with filters, rating and review system, and one-on-one tutoring session booking.

1.4 Scope and Limitation

1.4.1 Scope

The proposed system will be a fully responsive web application supporting two primary user roles: teachers and students. It will encompass key modules for course management, multimedia content delivery, assessments, payment processing, real-time communication, and AI-enhanced personalization features. A functional prototype will be developed, implemented, and tested with sample users.

1.4.2 Limitation

The system will be limited to a responsive web interface without native mobile applications. AI functionalities will rely on existing APIs and pre-trained models without custom training. Payment integration will operate only in test/sandbox mode. The project focuses on a working prototype rather than a fully scalable production system and does not include offline access or advanced accessibility features for differently-abled users.

1.5 Development Methodology

The project adopts an iterative Software Development Life Cycle (SDLC) approach incorporating Agile principles. It includes requirement gathering and feasibility study, system design using appropriate UML diagrams, implementation of the application, thorough unit and system testing, and comprehensive documentation. Progress will be monitored through regular supervisor consultations and maintenance of a logbook.

2 Problem Statement

The rapid expansion of online education in Nepal has created opportunities for flexible learning, yet the commercialization and accessibility of digital courses remain severely constrained by several practical and financial barriers [5, 7]. Most existing Learning Management Systems (LMS) and online education platforms available in Nepal either do not support local payment methods effectively or impose high transaction fees that make course monetization unattractive for teachers and expensive for students. These limitations significantly hinder the development of sustainable, locally-relevant online education ecosystems and restrict the ability of Nepali educators to generate income from digital content while keeping courses affordable for learners.

The major problems addressed by this project are:

- High commission fees (often 15–30%) charged by international payment gateways (e.g., PayPal, Stripe), which are either not fully available in Nepal or very costly for small-scale teachers and students [5, 7].
- Limited integration of popular and accessible local payment methods such as eSewa, Khalti, IME Pay, ConnectIPS, mobile banking, and Fonepay, making it difficult for the majority of Nepali users to enroll in paid courses conveniently [5].
- Absence of flexible pricing mechanisms, including coupon/discount code systems, dynamic price editing, and free demo/preview options, which are essential for attracting students and building trust in paid online courses [4].
- Lack of teacher-friendly monetization tools, such as easy course pricing management, performance-based revenue tracking, and support for both free and paid content models within the same platform [4].
- Insufficient interactive and value-added features (real-time chat, AI summarization, personalized recommendations, remedial suggestions, badges, quiz ranking, and one-on-one booking) that could justify paid enrollment and increase student retention in a competitive market [3, 4].
- Poor user experience due to missing advanced course discovery tools (smart search with filters, ratings/reviews) and limited support for multimedia delivery (video recording, live streaming with optional recording) tailored to local needs [4].

These financial, payment-related, and feature gaps discourage teachers from creating quality paid content, limit student participation in premium courses, and prevent the growth of

a vibrant, self-sustaining online education market in Nepal [5, 7]. This project proposes a modern web-based Learning Management System that addresses these specific challenges through seamless local payment integration, low-cost transaction design, flexible monetization options, and enhanced interactive features to make online education more accessible, affordable, and rewarding for both educators and learners.

3 Objectives

The primary objective of this project is to design, develop, and implement a modern, feature-rich, web-based Learning Management System (LMS) that addresses the key challenges of online education commercialization and accessibility in the Nepali context, making digital learning more affordable, interactive, and rewarding for both educators and students.

The specific objectives are as follows:

- To integrate secure and cost-effective local payment gateways (such as eSewa, Khalti, IME Pay, ConnectIPS, Fonepay, and mobile banking) with support for low or zero commission transactions to enable convenient and affordable course enrollment for Nepali users.
- To provide flexible monetization and pricing tools for teachers, including dynamic course price setting/editing, coupon/discount code generation, free preview/demo video uploads, and mixed free/paid content models to encourage quality content creation and teacher income generation.
- To implement advanced multimedia content delivery features, including video lecture recording and storage, live streaming with optional recording, and AI-powered automatic summarization of video lectures and textual notes (similar to advanced language models) to enhance learning efficiency.
- To develop interactive communication and engagement tools such as real-time group chat channels, discussion forums, and one-on-one online tutoring session booking to improve teacher-student and peer interaction.
- To incorporate AI-driven personalization and support features, including intelligent course recommendations based on user viewing history, cart behavior, and site visits; remedial content suggestions for struggling students; performance badges; and quiz ranking to increase student motivation and retention.
- To enable comprehensive teacher tools for course management, including syllabus upload, quiz/assignment creation and grading, student performance tracking, and content organization, while allowing students to preview syllabi, search/filter courses, rate and review courses, and access purchased materials seamlessly.
- To design a responsive, user-friendly web application suitable for the Nepali educational environment, ensuring ease of use on both desktop and mobile browsers with emphasis on accessibility and performance under typical local network conditions.

By achieving these objectives, the proposed LMS will contribute to building a more sustainable, inclusive, and effective online education ecosystem tailored to the needs and constraints of Nepal.

4 Methodology

The development of the proposed Learning Management System (LMS) follows a systematic approach based on the Software Development Life Cycle (SDLC) with iterative elements inspired by Agile methodology. This ensures continuous feedback, adaptability, and alignment with the requirements of the Nepali educational context. The methodology is divided into three main phases: Requirement Identification, Feasibility Study, and High-Level Design of the System.

4.0.1 Requirement Identification

Study of Existing System

A thorough study of existing Learning Management Systems and relevant literature was conducted to identify strengths, limitations, and opportunities for improvement in the Nepali context. Globally popular platforms such as Moodle, Canvas, Blackboard, and Google Classroom provide robust features for course management, assessments, and content delivery, but they often lack seamless integration with local Nepali payment gateways (e.g., eSewa, Khalti) and impose high customization costs or international transaction fees [3,4].

In Nepal, the shift to online education during COVID-19 relied heavily on tools like Zoom, Google Meet, and Microsoft Teams, with some institutions adopting basic LMS solutions [5,7]. However, these platforms suffer from poor support for local monetization, limited AI personalization, and inadequate features for multimedia interaction and student engagement [5]. Literature highlights the growing importance of AI-driven features such as content summarization using transformer models and recommendation systems based on vector similarity search to improve accessibility and retention [3,4]. The review confirms the need for a locally tailored LMS that integrates affordable payment methods, flexible pricing, real-time interaction, and AI enhancements while remaining cost-effective and user-friendly.

Requirement Analysis

Requirements were gathered through stakeholder consultation (teachers, students, and administrators), surveys of existing pain points, and analysis of literature. Functional requirements include: - Secure integration of local payment gateways with coupon/discount support - Course creation with flexible pricing, video upload, live streaming, quizzes, and assignments - AI-powered video/text summarization and personalized recommendations - Real-time group chat and one-on-one booking - Student features: search/filter, rating/review, remedial suggestions, and performance badges

Non-functional requirements emphasize responsiveness (mobile-first design), security (JWT authentication, HTTPS), performance (efficient under typical Nepali internet conditions), and usability (intuitive interface). Requirements were prioritized using the MoSCoW method to focus on core monetization and interaction features first.

4.0.2 Feasibility Study

Technical Feasibility

The project is technically viable using widely available open-source technologies. The backend will be developed with Node.js/Express.js and MongoDB for flexible data handling. Payment integration will use official APIs from eSewa, Khalti, and IME Pay (low commission rates of 1–3%). For AI features, pre-trained transformer models from Hugging Face (e.g., BART or T5 for summarization) will be employed via API calls or local inference on CPU/GPU. Recommendation system will utilize vector embeddings (e.g., text-embedding-3-large) combined with cosine similarity and approximate nearest neighbor search (using libraries like Faiss) [3,4]. Frontend will use React.js with Socket.io for real-time features. All technologies are within the scope of B.Sc.CSIT curriculum and can run on standard development hardware with cloud deployment options (Heroku/AWS free tier).

Operational Feasibility

The system aligns well with existing educational workflows in Nepal. Teachers can easily create and manage paid courses using familiar concepts, while students benefit from local payment methods and smartphone-friendly interface (smartphones being the primary device for many learners [5]). Minimal training is required due to intuitive design and built-in help sections. Maintenance will involve periodic updates to payment APIs and AI models, which can be handled by a small team or the developers themselves.

Economic Feasibility

Development costs are low as the project utilizes free/open-source tools (React, Node.js, MongoDB, Hugging Face models) with no licensing fees [3]. Hosting can start with free tiers and scale affordably. Local payment gateways offer significantly lower commissions than international alternatives, enabling sustainable monetization for teachers. The system aims to generate value through increased teacher income and student access to affordable quality education, with projected positive return on investment for a production version.

Table 4.1: Estimated Development Cost of the Proposed LMS

Cost Component	Estimated Cost (NPR)	Remarks
Software Tools	0	Open-source (Node.js, React, MongoDB)
Hardware	0	Existing personal computer
AI API key	0	Free API key from open router
Internet Charges	1,000	Development/testing phase
Hosting	5,000	VPS
Maintenance Cost	2,000	Minor future updates
Total Cost	8,000	Affordable and economical

Schedule (Gantt Chart Showing the Project Timeline)

The project is scheduled over 16 weeks (one academic semester):

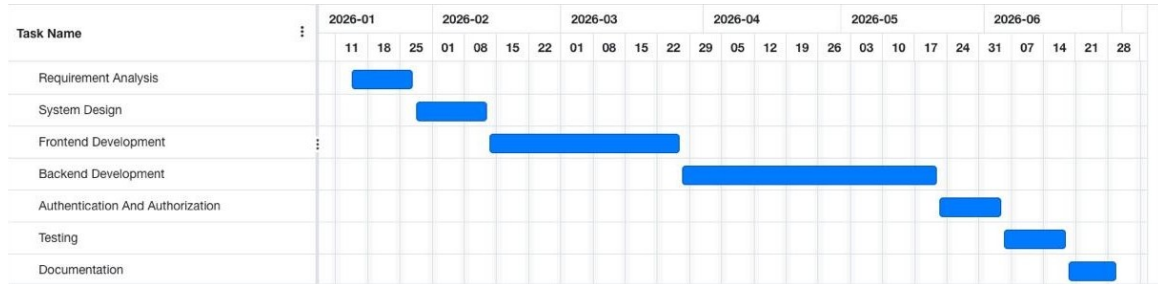


Figure 4.1: Gantt Chart Project Timeline

4.0.3 High Level Design of System (Methodology of the Proposed System / Working Mechanism / Description of Algorithms)

The proposed Learning Management System adopts a client-server architecture with a three-tier structure: Presentation Layer (React.js frontend), Application Layer (Node.js/Express.js backend), and Data Layer (MongoDB). This design ensures separation of concerns, scalability, and maintainability [3]. The frontend provides a responsive, mobile-friendly user interface, while the backend manages authentication (JWT), business logic, payment processing, and AI integrations. MongoDB handles persistent storage of users, courses, enrollments, payments, chat messages, and interaction logs.

4.0.4 Flowchart

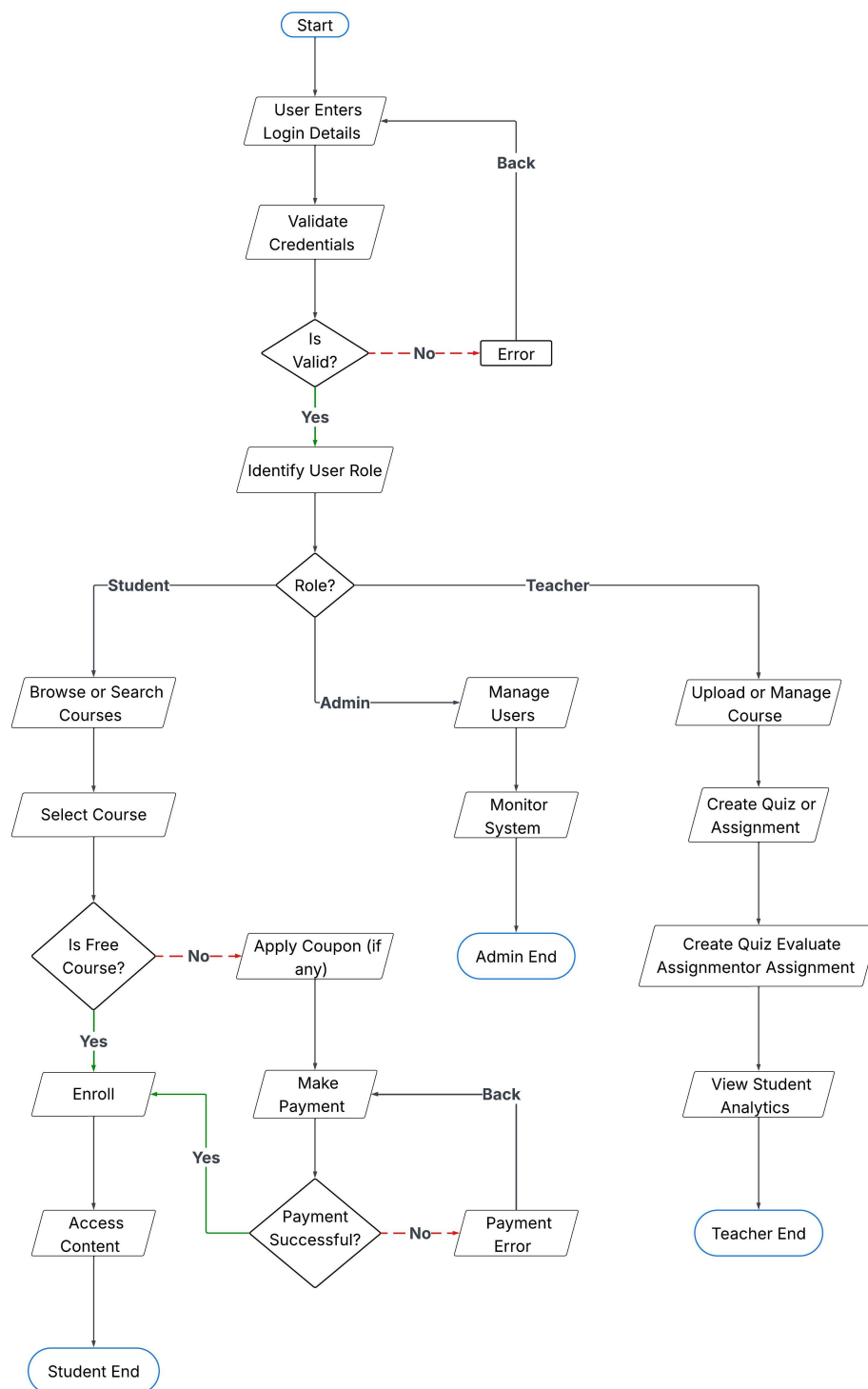


Figure 4.2: Flowchart

4.0.5 Use Case Diagram

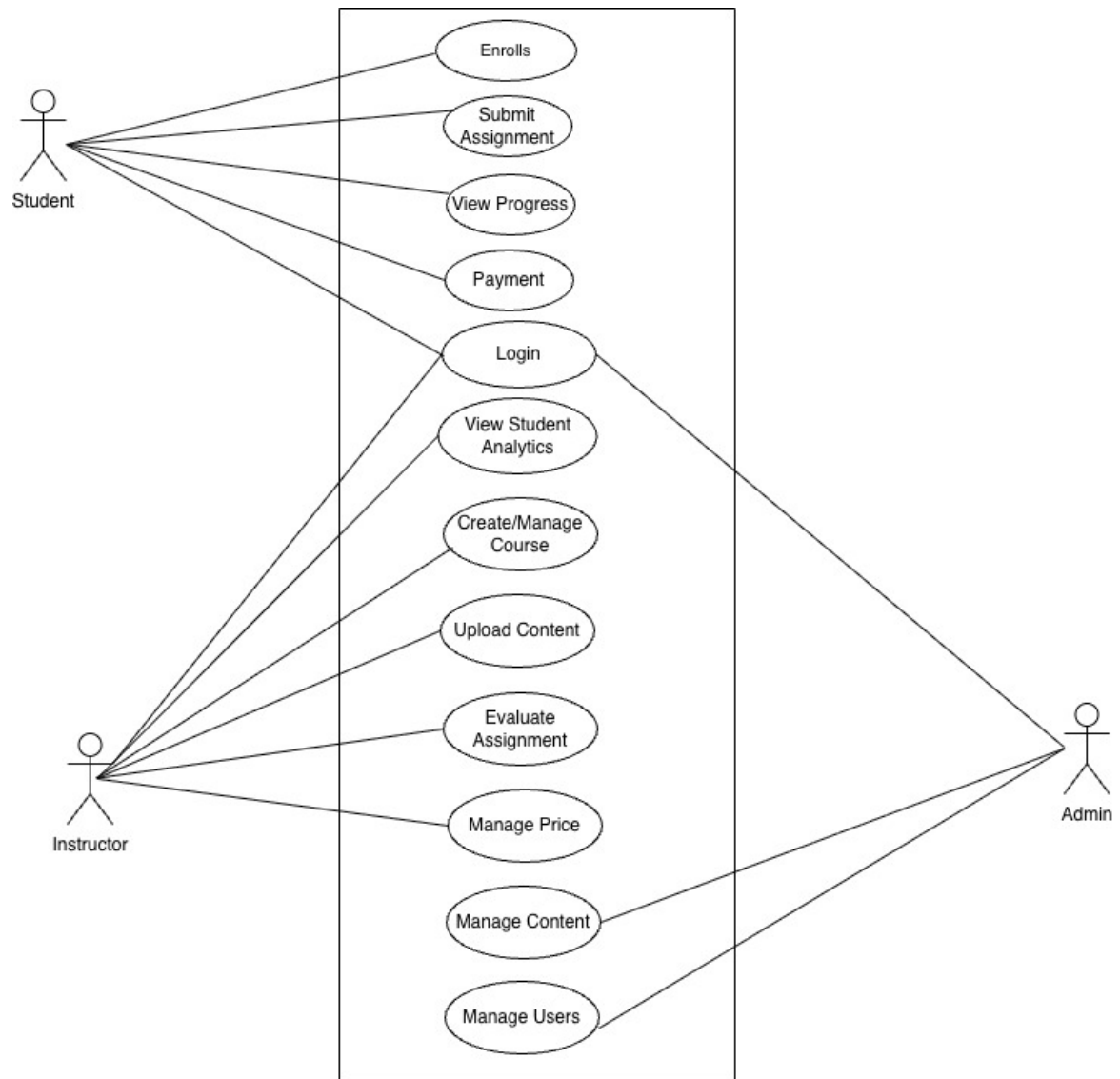


Figure 4.3: Use Case Diagram

4.0.6 Block Diagram

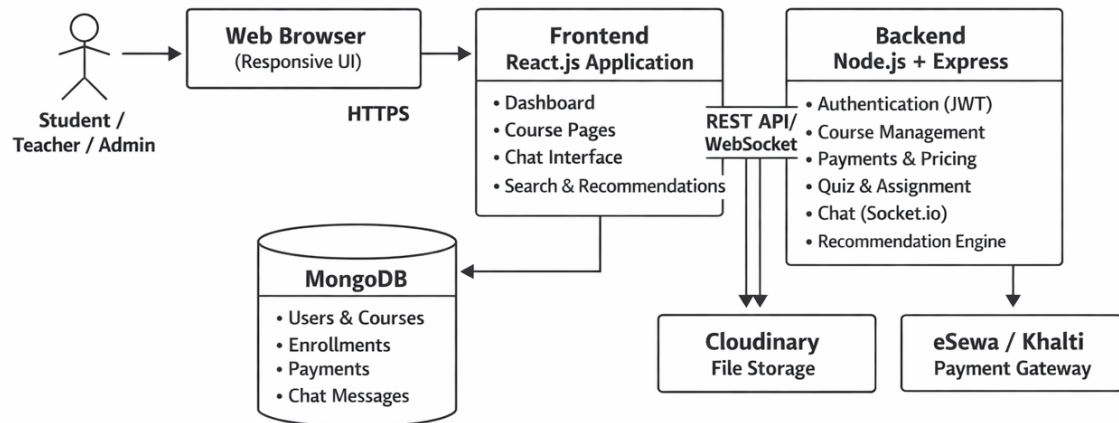


Figure 4.4: Block Diagram

4.0.7 Activity Diagram

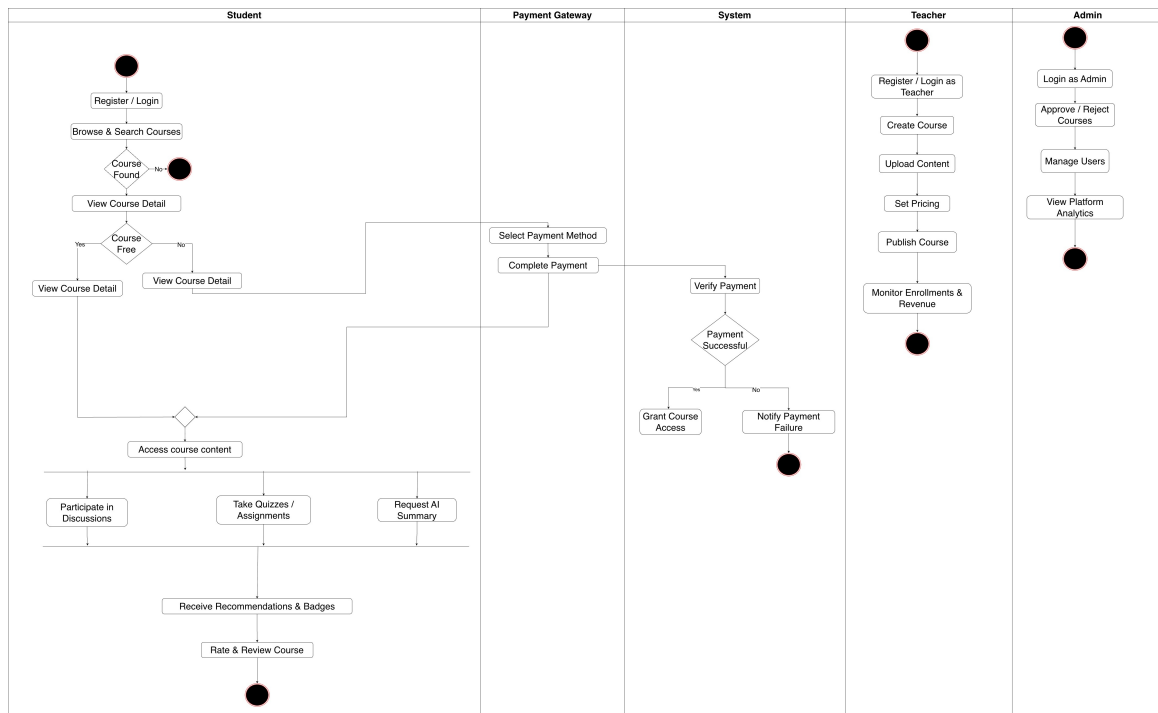


Figure 4.5: Activity Diagram

4.0.8 Sequence Diagram

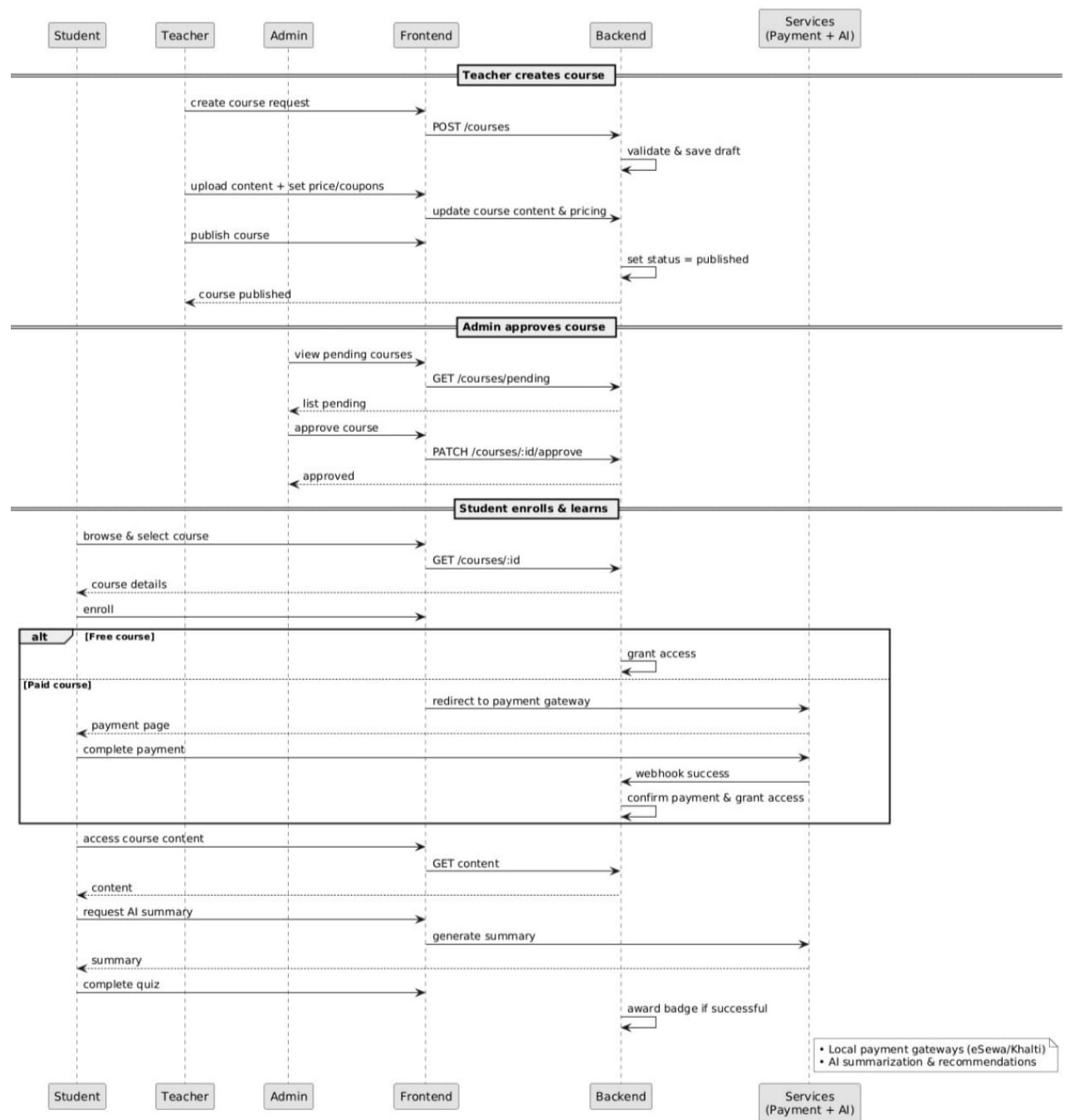


Figure 4.6: Sequence Diagram

Key Algorithms Employed:

- **Course Recommendation Algorithm (Content-based Filtering with Vector Search)**

The system uses **text-embedding-3-large** (OpenAI's high-performance embedding model) to create dense vector representations of both course content (title, description, syllabus, tags) and user interests (viewed courses, search history, cart items) [4].

Process:

1. Course metadata is embedded once during course creation/upload using text-embedding-3-large (dimensions: 3072).
2. User profile vector is dynamically computed as the weighted average of embeddings of recently interacted courses (recent views weighted higher).
3. Similarity between user profile vector $\mathbf{u}$ and course vector $\mathbf{c}$ is calculated using cosine similarity:

$$\text{cosinesimilarity}(\mathbf{u}, \mathbf{c}) = \frac{\mathbf{u} \cdot \mathbf{c}}{\|\mathbf{u}\| \|\mathbf{c}\|}$$

4. Top-k most similar courses (excluding already enrolled ones) are retrieved efficiently using an approximate nearest neighbor (ANN) index built with Faiss library (IVF or HNSW index), achieving sub-linear query time even for thousands of courses.

This approach provides highly relevant, personalized recommendations while leveraging the superior semantic understanding of modern embedding models.

- **Content Summarization Algorithm (Abstractive Summarization)**

For generating concise summaries of video transcripts (extracted via speech-to-text) or textual notes, a transformer-based encoder-decoder model is used (e.g., fine-tuned BART-large or facebook/bart-large-cnn from Hugging Face) [4].

Process:

1. Input text is tokenized and fed to the encoder.
2. The decoder autoregressively generates summary tokens based on encoder hidden states.
3. Beam search (beam width 4–6) is applied to improve summary quality.

The model produces abstractive, coherent summaries in near real-time, helping students quickly grasp key concepts even with limited time or connectivity.

- **Search Algorithm**

To provide personalized course recommendations, course embeddings are generated and compared with the user profile vector using cosine similarity. This helps students discover relevant courses based on their interests and previous activity.

Process:

1. Course Embedding is generated when a teacher creates or updates a course.

2. User Profile Vector is calculated by averaging embeddings of recently viewed/enrolled courses.
3. Similarity Search is performed using cosine similarity over all published courses.
4. Top-K recommendations are returned (e.g., top 5).

The algorithm improves user experience by providing personalized course suggestions, increasing engagement and helping students find relevant content faster.

These algorithms and mechanisms are implemented using proven, state-of-the-art tools (OpenAI text-embedding-3-large for embeddings, Hugging Face transformers for summarization, and Faiss for efficient vector search), making the system both powerful and feasible within the scope of a B.Sc.CSIT final year project [3, 4].

5 Expected Outcome

Upon successful completion of the project, **NEXL: Elevate Your Learning** will deliver a fully functional, full-stack web-based Learning Management System specifically tailored for the Nepali educational context. The expected outcomes are as follows:

- **A complete, deployable web application** that allows independent teachers and tutors to:
 - Create and manage courses using multiple content formats (video lectures, hand-written/scanned notes, and PDF documents)
 - Set flexible pricing structures, apply discount coupons, and edit them easily
 - Optionally provide free demo/preview videos to attract students
 - Create real-time chat groups and announcement channels for student-teacher interaction.
 - Design assignments and quizzes with automated evaluation features
 - Award performance-based digital badges to motivate students based on quiz scores, assignment completion, and class participation.
- **An intuitive student-facing platform** that enables users to:
 - Browse, search, and discover available courses efficiently
 - Receive basic personalized course recommendations based on interests and past interactions.
 - Securely purchase and enroll in paid courses using local payment gateways (eSewa/Khalti)
 - Access enrolled course content, participate in discussions, complete assessments, and earn badges
 - Submit ratings and detailed reviews to help improve course quality and platform credibility
- **Enhanced accessibility and affordability** for Nepali users by minimizing platform fees (low/no commission), supporting common local teaching formats (handwritten notes), and offering discount mechanisms suitable for students with varying economic backgrounds.
- **Positive impact potential** on the local e-learning ecosystem by empowering independent educators to generate sustainable income, providing students with motivating and interactive learning experiences, and contributing to the overall growth of digital education in Nepal.

The final system will be hosted on a cloud platform (initially free-tier Vercel/Netlify or affordable VPS), fully tested, documented, and ready for demonstration during the final defense. It will serve as a practical, scalable prototype that can be extended in the future (e.g., mobile app, advanced AI recommendations, more payment options).

Overall, NEXL is expected to become a useful, teacher-friendly, and student-centric alternative to existing global and local platforms, directly addressing the identified gaps in affordability, content flexibility, interaction, and motivation in Nepal's growing e-learning landscape.

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